

ANSIBLE – JENKINS DEPLOYMENT PROJECT

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Course: DevOps

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1. Project Overview

This project demonstrates automated deployment using Jenkins integrated with Ansible.

The workflow includes:

- Developer pushes code to Git repository
 - Jenkins pulls the code
 - Jenkins builds the project and generates WAR file
 - Jenkins triggers Ansible playbook
 - Ansible deploys the application to production server
 - Tomcat server runs the application
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Architecture / Workflow

1. Developer → Pushes code to Git
 2. Jenkins → Pulls the code
 3. Jenkins → Builds WAR file
(myapp.war)
 4. Jenkins → Triggers Ansible Playbook
 5. Ansible → Connects to Production Node
 6. Ansible → Copies WAR file to Tomcat
 7. Ansible → Restarts Tomcat
-

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Tools Used

- Jenkins
- Ansible
- Apache Tomcat
- Maven
- Git
- AWS EC2

Execution Screenshots

- EC2 Instances (Ansible + Prod Node)
- Jenkins Pipeline Stages
- Ansible Playbook Execution
- Deployment Output
- Final Application Output (Browser screenshot)

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Challenges Faced

- SSH key configuration issues
- Ansible connectivity problems
- Jenkins integration setup
- Deployment errors

Conclusion

This project successfully demonstrates automated deployment using Jenkins and Ansible. It reduces manual effort and ensures faster and reliable application deployment.

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The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The left sidebar shows the 'EC2' section with 'Instances' selected. The main content area displays 'Instances (2)' with a table listing two instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
ANSIBLE-JENK...	i-018d4c9b19085016b	Running	c7i-flex.large	3/3 checks passed	View alarms +
TOMCAT-SERV...	i-0394acbb625812881	Running	c7i-flex.large	3/3 checks passed	View alarms +

```

TOMCAT-SERVER
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
..
.ssh
.bashrc
.bash_profile
.bash_logout
Remote monitoring
Follow terminal folder

[ansible@ip-172-31-93-171 ec2-user]$ sudo visudo
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for ansible:
sudo: a password is required
[ansible@ip-172-31-93-171 ec2-user]$ exit
exit
[ec2-user@ip-172-31-93-171 ~]$ sudo visudo
[ec2-user@ip-172-31-93-171 ~]$ ls /root/tomcat/webapps/
ls: cannot access '/root/tomcat/webapps/': Permission denied
[ec2-user@ip-172-31-93-171 ~]$ ps -ef | grep tomcat
root      33129      1  0 09:16 ?        00:00:09 /usr/lib/jvm/java-17-amazon-corretto/bin/java -Djava.util.logging.config.
file=/root/tomcat/conf/logging.properties -Dorg.apache.catalina.security.SecurityListener.UMASK=0027 --add-opens=java.base/java.lang=ALL-UNNAMED --add
-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java
base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED --enable-native-access=ALL-UNNAMED -classpath /root/tomcat/bin/boo
tstrap.jar:/root/tomcat/bin/tomcat-juli.jar -Dcatalina.base=/root/tomcat -Dcatalina.home=/root/tomcat -Djava.io.tmpdir=/root/tomcat/temp org.apache.catali
na.startup.Bootstrap start
ec2-user  35291   34518  0 09:59 pts/0    00:00:00 grep --color=auto tomcat
[ec2-user@ip-172-31-93-171 ~]$ ^C
[ec2-user@ip-172-31-93-171 ~]$ sudo ls /root/tomcat/webapps/
ROOT docs examples host-manager manager myapp myapp.war
[ec2-user@ip-172-31-93-171 ~]$ ^C
[ec2-user@ip-172-31-93-171 ~]$
```

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← → ↻ ⚠ Not secure 13.221.236.167:8080 ☆ 🔍 👤

Jenkins 🔍 ⚙️ 👤

+ New Item

📁 Build History

Build Queue ▾
No builds in the queue.

Build Executor Status 0/2 ▾

All +

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☁	Ansible-project	30 min #3	32 min #2	22 sec ▶

Icon: S M L ...

← → ↻ ⚠ Not secure 13.221.236.167:8080/job/Ansible-project/ ☆ 🔍 👤

Jenkins / Ansible-project 🔍 ⚙️ 👤

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑 Delete Pipeline

🔍 Full Stage View

📁 Stages

✎ Rename

❓ Pipeline Syntax

Builds > ...

🔍 Filter /

Today
13.221.236.167:8080/job/Ansible-project/workflow-stage

✓ Ansible-project

Add description

Stage View

Average stage times:
(full run time: ~12s)

	Checkout	Compile-project	Test-project	Artifactory-file
#3 Apr 08 15:00 No Changes	246ms	10s	4s	6s
#1 Apr 08 14:54 No Changes	555ms			

Permalinks

- Last build (#3), 30 min ago

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← → ↻ ⚠ Not secure 44.201.227.63:8080/manager/html ☆ 🔍 🧑



Tomcat Web Application Manager

Message: OK

Manager			
List Applications	HTML Manager Help	Manager Help	Server Status

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

← ↻ ⚠ Not secure 44.201.227.63:8080/myapp/ Summarize 🔍 ☆ ⚙️ 👤 Chat

Disney+ hotstar TV Movies Sports Premium Disney+ KIDS

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Luca

Lorem ipsum dolor sit amet consectetur adipisicing elit. Velit porro et veniam excepturi, eaque voluptatem impedit nulla laboriosam facilis ut laboriosam libero!